

Uttkarsh Tyagi

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Data, cloud and ML engineer who has shipped real, running systems: a live AWS platform with a PyTorch model serving predictions in production, a serverless medallion data lake, and a LangGraph chatbot whose every number is verified. Recently completed **C-DAC's PG Certificate Programme in Big Data Analytics (PGCP-BDA)**. Comfortable picking up new tools fast to fit a stack.

EDUCATION

C-DAC, Mumbai — PG Certificate Programme in Big Data Analytics (PGCP-BDA)

Feb 2026 – Sep 2026

Grade A · 72.71%

Uttarakhand Technical University, Dehradun

Sep 2021 – Jul 2025

B.Tech, Information Technology · CGPA 6.5

Delhi Public School, Ranipur, Haridwar

Class 12 — 88.4% · Class 10 — 86%

SKILLS

Data Engineering	ETL/ELT pipelines, medallion lake (bronze/silver/gold), DuckDB, Apache Spark, Kafka/Redpanda, Delta Lake, dbt, Apache Airflow, data-quality and drift checks
Cloud (AWS)	Lambda, S3, API Gateway, EventBridge, DynamoDB, ECR, IAM (GitHub OIDC), CloudWatch, Terraform (IaC), GitHub Actions (CI/CD), Docker, Linux
ML / GenAI	PyTorch (trained + deployed to production), ONNX, scikit-learn, MLflow, anomaly detection and forecasting; LangGraph tool-calling chatbots, LLM APIs (Groq, Gemini), RAG with embeddings and vector search (earlier StreamPulse version)
Backend & APIs Languages	Python (NumPy, pandas), FastAPI, REST API design, Next.js/TypeScript, PostgreSQL, SQL, DynamoDB Python, SQL, TypeScript/JavaScript, Bash

PROJECTS

liveflights — Real-time flight intelligence platform

github.com/tyxgx/liveflights · [live demo](#)

AWS Lambda/S3/API Gateway/CloudWatch · Terraform · FastAPI · Next.js · PyTorch · ONNX · Airflow

- A real-time platform that tracks live aircraft over Europe: a scheduled AWS Lambda pulls live flight positions every minute into S3, and a FastAPI backend serves them to a Next.js map dashboard showing ~3,500 aircraft from 50+ countries.
- The whole system is defined in Terraform and runs serverless (Lambda, S3, API Gateway, EventBridge, Firehose) for ~\$2/month, with CloudWatch alarms for missing data, errors and failed deliveries.
- ML on the live feed: a PyTorch GRU predicts each aircraft's position 5 minutes ahead (served from Lambda via ONNX) and every prediction is scored against the real position later (median error ~1-1.5 km); also anomaly detection and an hourly traffic forecast (MAE 86 vs 568 for a naive baseline).
- Debugged real production issues: batched ONNX inference to fix a Lambda timeout, lazy loading to fix an out-of-memory error, and a plausibility guard after impossible speeds exposed aircraft-address collisions in the feed.
- Data-quality and drift checks, compaction, dbt and model retraining run as 4 Airflow DAGs, tested end to end in GitHub Actions CI.

StreamPulse — Serverless data platform + dashboard

github.com/tyxgx/steampulse · [live demo](#)

DuckDB · AWS S3/Lambda/DynamoDB · Terraform · GitHub Actions · LangGraph · Chart.js/D3

- A Spotify charts analytics platform: daily top-200 charts for 72 markets (44.6M rows since 2017) turned into a dashboard (home + 9 views) and a chatbot you can ask questions.
- A GitHub Actions job runs daily with DuckDB: raw data goes into an append-only Bronze layer with change detection (row hashes), a cleaned Silver layer is rebuilt from Bronze, and aggregated Gold tables land on S3, with a Gold-vs-Silver check on every run. Infrastructure is Terraform, with GitHub OIDC instead of stored AWS keys.
- The dashboard is static pages built from ~8 MB of precomputed JSON, so visitors trigger no queries; a data-health page shows freshness, row counts and null checks.
- The chatbot is a LangGraph agent on AWS Lambda: 10 deterministic DuckDB tools produce every number, the model only writes the answer, and a verifier rejects any figure a tool did not return (38/38 on a golden set).
- Found and fixed a non-deterministic aggregation that changed artist counts between runs, and covered the pipeline with 11 tests (incremental load matches a full rebuild, reruns are idempotent).

Interactive ML — Stage-by-stage ML pipeline explorer

github.com/tyxgx/interactive-ml · [live demo](#)

FastAPI · scikit-learn · Pydantic · pytest · Next.js · TypeScript

- Built an ML pipeline explorer (load, split, preprocess, train, evaluate as inspectable steps) with leakage-safe preprocessing; the FastAPI backend runs in the browser via Pydantic, with 38 pytest tests and CI.

EXPERIENCE

Full Stack Developer Intern — *Unlock Discounts, Bangalore (Remote)*

Jul 2025 – Oct 2025

- Built a customer review section (React/JS/CSS) and APIs to fetch and display data from the company's database across platforms; created a certificate verification system for intern authentication.

Project Intern — *BHEL, Haridwar*

Jul 2025 – Aug 2025

- Developed a web portal (HTML/CSS/React, SQL backend) for equipment issuance with late tracking and recipient records.